

EMG Chatbot — Team Brief

AUT ENSE891 | Signal-Driven AI Systems | 15 June 2026

What we are building

A chatbot where users ask questions about EMG fatigue signals. The system shows an interactive Plotly signal panel alongside a text answer, grounded by the pipeline's real classifier output — no image guessing.

Three roles — one seam

Rayyan

`render_window() + build_prompt()`

Visual panel + LLM grounding bridge

Branch: `feat/rayyan-visualization` Folder: `viz/`

Aryan

`classify()`

Transformer / LSTM fatigue features

Branch: `feat/aryan-ml` Folder: `models/`

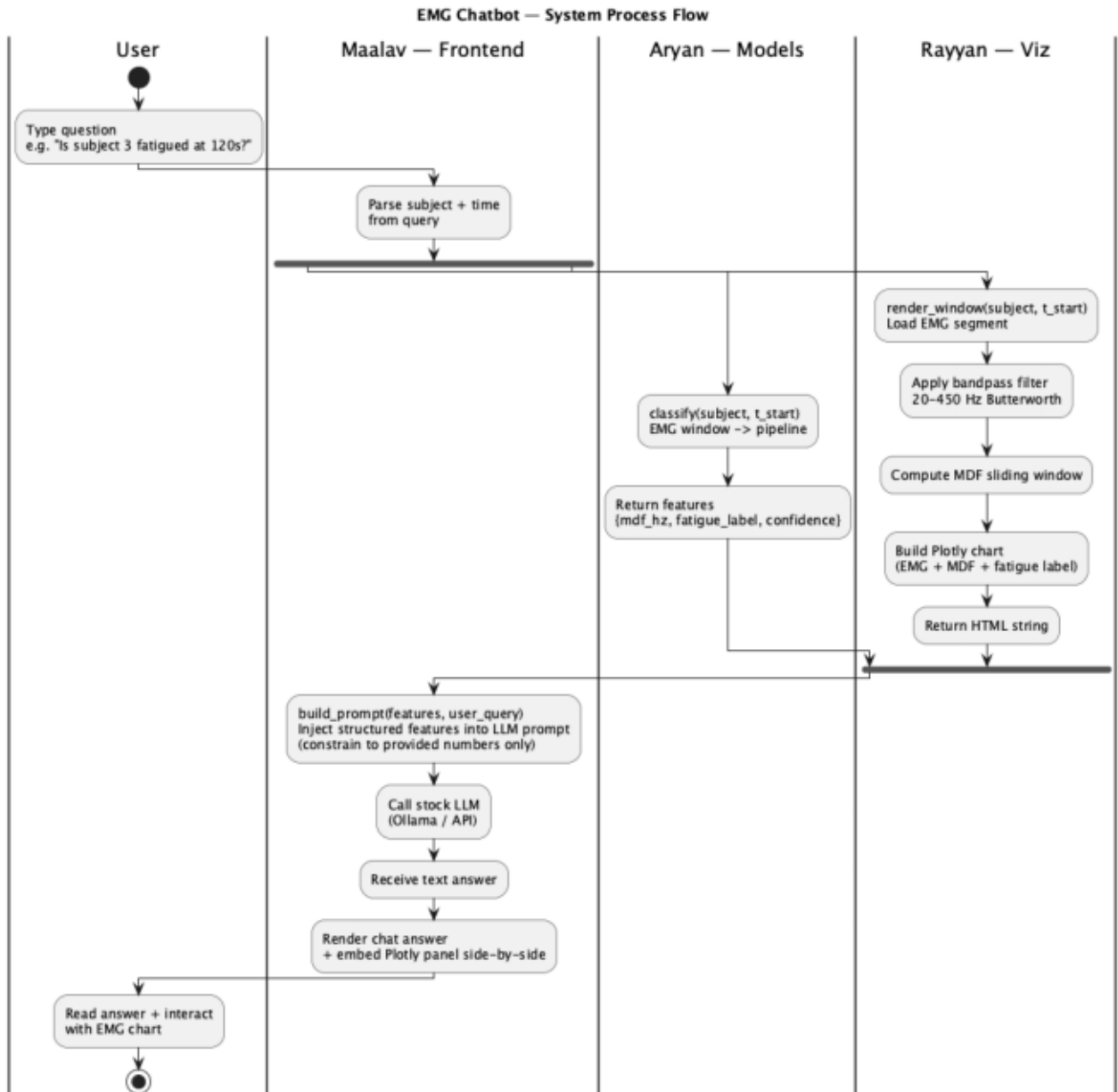
Maalav

Chat UI + LLM wiring

Frontend, prompt injection, display

Branch: `feat/maalav-frontend` Folder: `frontend/`

System Process Flow — swimlane by owner



Integration Contract — agree signatures before building

Function	Signature	Returns	By	For
render_window	render_window(subject: int, t_start: float, side: str = 'R')	str — Plotly HTML (no full_html wrapper) OR fig.to_json() if Maalav uses React+plotly.js	Rayyan	Maalav
classify	classify(subject: int, t_start: float, side: str = 'R')	dict: mdf_hz: float fatigue_label: int confidence: float	Aryan	Maalav
build_prompt	build_prompt(features: dict, user_query: str)	str — LLM system prompt constraining model to provided numbers only	Rayyan	Maalav

Maalav: confirm Python (Streamlit/Gradio) vs JS/React first — determines render_window return type.

Git Workflow + Getting Started

Branch rules

main	stable – merged only via PR, never push direct
feat/rayyan-visualization	Rayyan (already on remote)
feat/aryan-ml	Aryan creates: <code>git checkout -b feat/aryan-ml</code>
feat/maalav-frontend	Maalav creates: <code>git checkout -b feat/maalav-frontend</code>

Getting started

```
git clone https://github.com/gargmaalav/Final-Year-Project.git

git checkout main && git pull

git checkout -b feat/<your-branch>

# work in your folder (models/ or frontend/)

git add <files>

git commit -m 'feat(<scope>): description'

git push -u origin feat/<your-branch>

# open a Pull Request into main when your piece works end-to-end
```

Commit message format

```
feat(viz): add render_window Plotly chart

feat(ml): add LSTM fatigue classifier

feat(frontend): wire classify() into LLM prompt

fix(viz): correct MDF cursor x-position
```

Types: feat fix refactor docs test chore perf

Subject: lowercase, imperative mood, no trailing period, under 72 chars

Rayyan Abzal | AUT ENSE891 | github.com/gargmaalav/Final-Year-Project

Dataset

Zenodo 14182446 -> zenodo.org/records/14182446

Place at: `zenodo_biceps/sEMG_data/` (gitignored — do not commit the data)

13 subjects, biceps brachii sEMG, 1259 Hz, right + left arm trials